RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Tutorial 9B: Applications of Integration

Section A (Basic Questions)

1 It is given that the curve C has equation $y=\frac{1}{x}-2$ with $x>0$. Find

(i) the volume of the solid formed when the region A bounded by C, the x-axis and the line $x=1$ is rotated completely about the x-axis,

(ii) the volume of the solid formed when the region B bounded by C, the axes and the line $y=-1$ is rotated completely about the y-axis.

The region bounded by C and the line $y=1-2x$ is denoted by R. Find

(iii) the area of R,

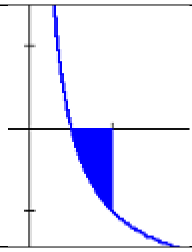
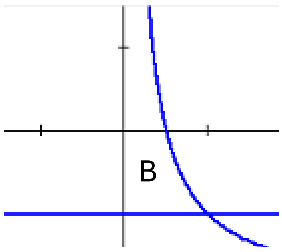
(iv) the volume of the solid formed when R is rotated completely about the x-axis,

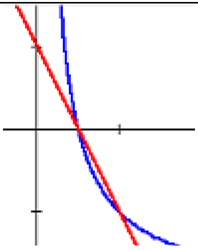
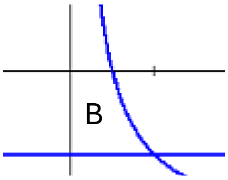
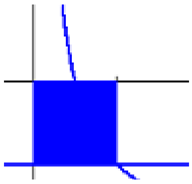
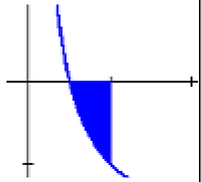
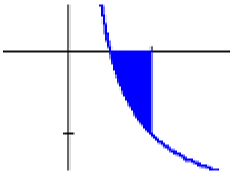
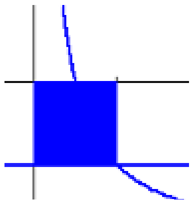
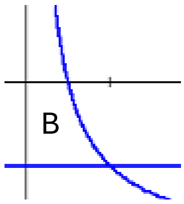
(v) the volume of the solid formed when R is rotated completely about the y-axis,

(vi) the volume of the solid formed when B is rotated completely about the x-axis,

(vii) the volume of the solid formed when A is rotated completely about the y-axis.

[**(i)** 0.714, **(ii)** 1.57, **(iii)** 0.0569, **(iv)** 0.191, **(v)** 0.262, **(vi)** 2.43, **(vii)** 1.57]

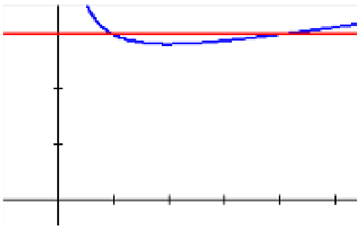
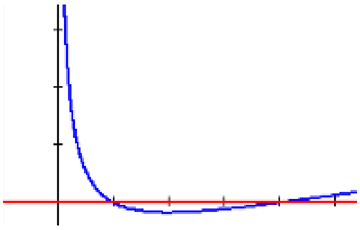
Solution:	
(i)	The curve meets the x-axis at $x=\frac{1}{2}$. Volume $=\pi\int_{\frac{1}{2}}^{1} y^2 \, dx = \pi\int_{\frac{1}{2}}^{1}\left(\frac{1}{x}-2\right)^2 dx = 0.714$ (3sf) 
(ii)	$y=\frac{1}{x}-2 \Rightarrow x=\frac{1}{y+2}$ Volume $=\pi\int_{-1}^{0} x^2 \, dy = \pi\int_{-1}^{0}\left(\frac{1}{y+2}\right)^2 dy = 1.57$ (3sf) 

(iii)	 The points of intersection are $\left(\frac{1}{2},\, 0\right)$ and $(1,\, -1)$. Area $= \int_{\frac{1}{2}}^{1} (1-2x) - \left(\frac{1}{x}-2\right) \, \mathrm{d}x = 0.0569 \ \text{(3sf)}$	To find the area between curves, use $\int_a^b \left(\mathrm{f}_1(x) - \mathrm{f}_2(x)\right) \mathrm{d}x$ where $\mathrm{f}_1(x) \geq \mathrm{f}_2(x)$. Details on page 9 of the notes.
(iv)	Volume $= \pi \int_{\frac{1}{2}}^{1} \left(\frac{1}{x}-2\right)^2 \mathrm{d}x \ - \ \pi \int_{\frac{1}{2}}^{1} (1-2x)^2 \ \mathrm{d}x$ $= 0.191 \ \text{(3sf)}$	Volume of revolution when area between curve and x-axis is rotated about x-axis Minus Volume of revolution when area between line and x-axis is rotated about x-axis. Details on page 13 of the notes.
(v)	$y = 1-2x \ \Rightarrow \ x = \frac{1-y}{2}$ Volume $= \pi \int_{-1}^{0} \left(\frac{1-y}{2}\right)^2 \mathrm{d}y \ - \ \pi \int_{-1}^{0} \left(\frac{1}{y+2}\right)^2 \mathrm{d}y$ $= 0.262$ (3sf)	Volume of revolution when area between line and y-axis is rotated about y-axis Minus Volume of revolution when area between curve and y-axis is rotated about y-axis. Details on page 13 of the notes.
(vi)	 Volume $= \pi \int_0^1 (y_1)^2 \, \mathrm{d}x \ - \ \pi \int_{\frac{1}{2}}^{1} (y_2)^2 \mathrm{d}x$ $= \pi \int_0^1 (-1)^2 \, \mathrm{d}x \ - \ \pi \int_{\frac{1}{2}}^{1} \left(\frac{1}{x}-2\right)^2 \mathrm{d}x$ $= 2.43$ (3sf)	Volume of revolution when area between the line $y = -1$ and x-axis is rotated about x-axis  Minus Volume of revolution when region A between curve and x-axis is rotated about x-axis. 
(vii)	 Volume $= \pi \int_{-1}^{0} (x_1)^2 \, \mathrm{d}y \ - \ \pi \int_{-1}^{0} (x_2)^2 \mathrm{d}y$ $= \pi \int_{-1}^{0} 1^2 \, \mathrm{d}y \ - \ \pi \int_{-1}^{0} \left(\frac{1}{y+2}\right)^2 \mathrm{d}y$ $= 1.57$ (3sf)	Volume of revolution when area between the line $x = 1$ and y-axis is rotated about y-axis  Minus Volume of revolution when region B between curve and y-axis is rotated about y-axis. 

2 The region R is bounded by the curve C with equation $y=\sqrt{x}+\dfrac{2}{\sqrt{x}}$ and the line $y=3$.

(i) Write down the equation of the curve obtained when C is translated 3 units in the negative y–direction. [1]

(ii) Hence, show that the volume of the solid formed when R is rotated completely about the line $y=3$ is given by $\pi\int_1^4\left(x-6\sqrt{x}+13-\dfrac{12}{\sqrt{x}}+\dfrac{4}{x}\right)\mathrm{d}x$. [4]

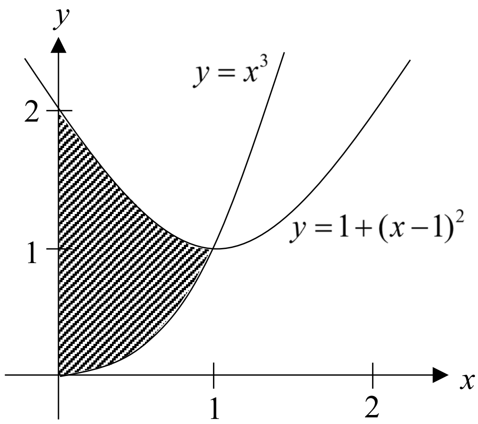
9205/J1994/01/Q16 (Modified)		
Solution:		
(i)	$y=\sqrt{x}+\dfrac{2}{\sqrt{x}}-3$	Replace y in the equation of curve C with $y+3$
(ii)	 	The required volume is the same as the volume of the solid obtained when the region between the curve in part (i) and the x-axis is rotated completely about the x-axis.
	Volume $=\pi\int_1^4 y^2\,\mathrm{d}x$ $=\pi\int_1^4\left(\sqrt{x}+\dfrac{2}{\sqrt{x}}-3\right)^2\mathrm{d}x$ $=\pi\int_1^4\left[x+2\sqrt{x}\left(\dfrac{2}{\sqrt{x}}-3\right)+\left(\dfrac{2}{\sqrt{x}}-3\right)^2\right]\mathrm{d}x$ $=\pi\int_1^4\left(x+4-6\sqrt{x}+\dfrac{4}{x}-\dfrac{12}{\sqrt{x}}+9\right)\mathrm{d}x=\pi\int_1^4\left(x-6\sqrt{x}+13-\dfrac{12}{\sqrt{x}}+\dfrac{4}{x}\right)\mathrm{d}x$ (shown)	

3 With reference to the given figure,

(i) find the exact area of the shaded region.

(ii) find the exact volume generated by the shaded region, when it is rotated through 4 right angles about the y-axis.

[(i) $\dfrac{13}{12}$, (ii) $\dfrac{23}{30}\pi$]



RJC Prelims 9233/2004/02/Q4
Solution:

By observation, the curves intersect at (1, 1).

(i)

Area of shaded region $= \int_0^1 \left[1+(x-1)^2 - x^3\right] \, \mathrm{d}x$

$$= \left[x + \frac{(x-1)^3}{3} - \frac{x^4}{4} \right]_0^1$$

$$= 1 - \frac{1}{4} + \frac{1}{3} = \frac{13}{12} \text{ units}^2$$

Alternatively, Area $= \int_1^2 1 - \sqrt{y-1} \, \mathrm{d}y + \int_0^1 y^{\frac{1}{3}} \, \mathrm{d}y$

To find the area between curves, use $\int_a^b \left(\mathrm{f}_1(x) - \mathrm{f}_2(x)\right) \mathrm{d}x$ where $\mathrm{f}_1(x) \geq \mathrm{f}_2(x)$. Details on page 9 of the lecture notes.
The second method considers the region as the area bounded by the curves and the y-axis. The region with $1 < y < 2$ is bounded by the left branch of the parabola, while the region with $0 < y < 1$ is bounded by the other curve.

(ii)

The parabola has equation

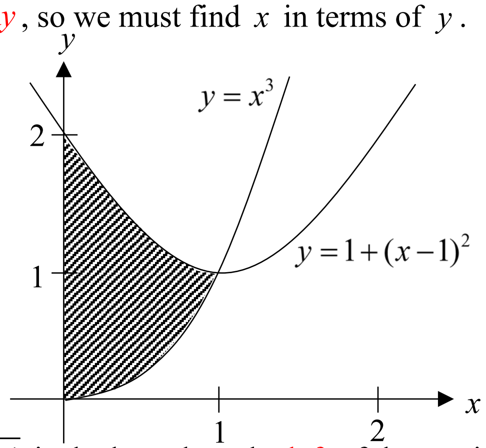
$$y = 1 + (x-1)^2$$

$$(x-1)^2 = y - 1$$

$$x = 1 \pm \sqrt{y-1}$$

The shaded region from $y = 1$ to $y = 2$ is bounded by $x = 1 - \sqrt{y-1}$

Since the volume is obtained by rotating the region about the y-axis, we will use $\pi \int_c^d x^2 \, \mathrm{d}y$, so we must find x in terms of y.



Note that $x = 1 - \sqrt{y-1}$ is the branch to the left of the turning point at $x = 1$, while $x = 1 + \sqrt{y-1}$ is the branch to the right of the turning point.

Volume generated by the shaded region, when it is rotated about the y-axis

$$= \pi \int_0^1 \left(y^{\frac{1}{3}} \right)^2 \mathrm{d}y + \pi \int_1^2 \left(1 - \sqrt{y-1}\right)^2 \mathrm{d}y$$

$$= \pi \left[\frac{y^{\frac{5}{3}}}{\frac{5}{3}} \right]_0^1 + \pi \int_1^2 \left[1 - 2\sqrt{y-1} + (y-1) \right] \mathrm{d}y$$

$$= \frac{3}{5}\pi + \pi \int_1^2 \left(y - 2\sqrt{y-1} \right) \mathrm{d}y$$

$$= \frac{3}{5}\pi + \pi \left[\frac{y^2}{2} - \frac{2(y-1)^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^2$$

$$= \frac{3}{5}\pi + \pi \left[\frac{4-1}{2} - \frac{4}{3}(1-0) \right] = \frac{23}{30}\pi \text{ units}^3$$